European Economic Review 37 (1993) 861–890. North-Holland

Productivity growth and the structure of the business cycle*

Gilles Saint-Paul

DELTA (Joint research unit ENS-CNRS-EHESS), Paris, France, CERAS, Paris, France and CEPR, London, UK

Over the recent years 'opportunity cost' (OC) models of growth have argued that recessions are times when firms engage in productivity-improving activities because of intertemporal substitution. This paper tests whether this approach is correct. The technique used to disentangle the trend from the cycle is semi-structural vector autoregression. The results are mildly supportive of the OC theory. There tends to be a negative impact of demand shocks on productivity, both in the short-run and in the long-run. And the short-run impact is stronger in those countries where fluctuations are more transitory. However, there is no significant response of R & D to demand shocks.

1. Introduction

This paper investigates the interactions between the business cycle and productivity growth. The two phenomena have traditionally been studied separately: business cycle theory was independent from and inconsistent with long-run growth theory and business cycle theorists used to study detrended data and to consider the trend as exogenous to the cycle. In the eighties, two strands of literature have emerged which relate the two phenomena. The real business cycle literature has suggested that part or all of what is called the business cycle may in fact be due to responses to productivity shocks. The endogenous growth literature emphasizes that long-run growth has numerous determinants, and in particular that transitory disturbances may have long-run effects on productivity growth. This leaves scope for a causal relationship from the cycle to the trend. The purpose of this paper is to empirically investigate such a relationship.

The starting point of my analysis is a class of models hereafter labeled as the 'opportunity cost' (OC) approach to productivity growth. These models insist on intertemporal substitution of productivity-improving activities along

Correspondence to: Gilles Saint-Paul, DELTA, 48 Boulevard Jourdan, 75014 Paris, France.

*I thank Philippe Aghion, Antoine d'Autume, Tamim Bayoumi, Charlie Bean, Martin Feldstein, Robert Gordon, Mervyn King, Jacques Mairesse and Georges de Ménil for helpful comments and suggestions, as well as Geneviève Muzart for providing data and Marie-José Pérez for technical assistance.

the business cycle. They typically predict that recessions are associated with a higher pace of productivity improving activities.[1] The aim of this paper is to test whether this implication is true, whether it is due to an intertemporal substitution effect, and whether research and development may be included in the list of activities which are governed by such an effect.

For this, I use the 'semi-structural' VAR methodology, as pioneered by Blanchard and Quah (1989). It consists in using assumptions based on economic theory and/or common sense in order to recover structural innovations from VAR innovations.

As will be made clear below, the results provide mild support for the OC approach. On the other hand, they say little on the response of R & D activities to business cycles: this is due to the lack of significance of the regression results which include R & D variables.

The paper is organized as follows; section 2 discusses the 'OC' approach to productivity growth. Section 3 describes the empirical methodology. Section 4 presents the main regression results. Section 5 discusses the implications for the trend vs. cycle decomposition. Section 6 presents the results on R & D. Section 7 contains concluding comments.

2. The 'opportunity cost' theory of productivity growth

A number of papers have recently emphasized that more activity may be going on during recessions than is traditionally believed. Moreover, some of these activities may play quite an important role in improving productivity. This literature has made both theoretical and empirical advances. A common theme is that productivity-improving activities are more important during recessions as a result of intertemporal substitution between these activities and directly productive activities: since the return to directly productive activities is lower in recessions due to lower demand, the opportunity cost in terms of foregone profits of such activities is lower in recessions than in expansions. This is why I will refer to these contributions as the 'opportunity cost' approach to productivity growth. I now briefly review some of this literature.

Davis and Haltiwanger (1989) found that more labor reallocation is going on during recessions. They propose a theoretical model according to which recessions are the appropriate time for labor reallocation as a result of intertemporal substitution.

Bean (1990) argues that such productivity-improving activities being more important in recessions may explain the pro-cyclical behaviour of productivity: if firms allocate a larger share of their labor force to such activities during recessions, then the actual input into directly productive activities goes down by more than the observed labor input, so that measured total

[1] These ideas go back at least to Schumpeter (1939, 1950).

factor productivity goes down. An important aspect of Bean's work is the long-run negative effect of a demand shock on productivity, which he establishes using a vector autoregression. This result is supportive of the OC theory.

Gali and Hammour (1991), which I follow in the empirical work below, use a semi-structural VAR approach in the vein of Blanchard and Quah (1989) and replicate Bean's result of a negative long-run effect of demand shocks on productivity.[2] They argue that within-firm training activities are more important during recessions and propose a theoretical model which supports this view. Their theoretical model also rests on intertemporal substitution effects.

Hall (1991) argues that recessions are periods of reorganization and introduces the concept of 'organizational capital'. He then proposes a model which implies that more organizational capital accumulation is going on during recessions.

Caballero and Hammour (1991) study a vintage capital model of investment and show that the firm will scrap old capital during recessions. They calibrate the model, showing that it is possible to replicate the Davis and Haltiwanger results.

Aghion and Saint-Paul (1991) develop a two-state Markov model of recessions and expansions. At each instant of time the firm has to decide the speed at which it moves along a 'technological axis'. The faster the firm moves, the higher the share of profits it loses. This captures the disruptive effects of restructuring and implementation of new technologies. As a result the opportunity cost of productivity-improving activities goes down in recessions because the cost is borne during the recession whereas the benefits are spread throughout the future. They study the effects on long-run growth of the amplitude and frequency of fluctuations. They argue that a larger amplitude is likely to have positive effects on long-run growth, and that long-run growth goes first up and then down when the frequency of recessions goes up, going through a 'Golden Rule' maximum where recessions are exactly as frequent as expansions.

Last, in Saint-Paul (1991), I provide tentative evidence that countries which had a higher unemployment rate in the post-1974 period also had a higher rate of total factor productivity growth. This suggests that the crisis following the first oil shock increased the pace of restructuring activities, more so in countries where it was deeper.

Criticisms and other stories

Three questions emerge when one comes to evaluate the validity of OC models.

[2]See also Caballero's (1993) comment on Bean and Pissarides, which provides additional supportive evidence.

1. Why is it that the productivity-improving activities one is talking about are different from traditional investment activities? If they are not different, why is it that the former should be 'counter-cyclical' whereas the latter are typically believed to be 'pro-cyclical'? Why is it that capital accumulation should react differently according to which type of 'capital' one is talking about? The size of this work does not allow to answer such a complex question, which is a research program in itself. I will just mention two potential explanatory factors.

First, one may depart from the 'Keynesian' view that recessions are associated with a waste of resources and consider that (part of) the seemingly unemployed resources are in fact used in activities which may not show up directly in national accounts data. This implies that there *must* be some counter-cyclical activities. Some of the investment activities may therefore be counter-cyclical while others are pro-cyclical. The split between the two will be determined by the *horizon* over which the benefits of the investment activities are reaped. Those types of investment which yield benefits for a longer time are more likely to be counter-cyclical because the firm is more willing to wait for the next expansion in order to benefit from them. On the contrary, investments which increase profits only over a short period of time (i.e., which depreciate more quickly) are likely to be pro-cyclical because firms want to use them at times of high demand. Organizational and knowledge capital probably depreciate less rapidly than physical capital. It is therefore reasonable to hypothesize that investment in the former may be counter-cyclical while physical investment is pro-cyclical.

Second, one may think that financial constraints are important in limiting the firms' access to new resources. This implies that cash-intensive investment activities are likely to be pro-cyclical, while other investment activities may be counter-cyclical. Therefore, investments associated with an inflow of material resources will probably be pro-cyclical, while investments based on a redistribution of existing resources across alternative activities will rather be counter-cyclical.

2. Even if one believes in the existence of such effects, one may question their empirical relevance; after all, one may think of many effects which would go in the opposite direction: learning by doing automatically introduces a positive feedback from activity to productivity growth; demand spillovers [Shleifer (1986)] may well induce firms to implement new technologies in expansion rather than in recessions, etc.; furthermore, these effects are more familiar and more often mentioned in the literature than intertemporal substitution effects.[3]

3. Even if one is ready to believe that recessions may have positive effects

[3]See Stadler (1990), Ambler et al. (1991). Bean and Pissarides (1993) study the impact of unemployment on growth through savings in a model with constant returns to capital.

on productivity, this does not imply that these effects have to do with intertemporal substitution; one may think of many other stories:

– 'Lame duck' effects: recessions eliminate the least productive firms thus freeing resources for the most productive firms.
– 'Discipline' effects: recessions put more firms in jeopardy by increasing the likelihood of bankruptcy, thus prompting them to reorganization efforts, etc. Increase in indebtedness may also have discipline effects [see Nickell et al. (1992)]. Aghion and Saint-Paul (1991) get this type of effect in the context of their model by introducing fixed costs.
– 'Externality effects': if one believes in the existence of a positive feedback from the *average* quality of labor to productivity growth, then recessions may have a positive impact on productivity because unskilled workers suffer more from them than skilled workers [see Dellas (1991)]. King and Robson (1990) study the long-run effects of non-economic disturbances (wars, etc.) and get similar predictions in a 'learning-by-watching' model where the investment rate affects productivity through an externality.

These effects are not incompatible with intertemporal substitution effects, but it may be difficult to distinguish them empirically.

In some sense, the present study tries to bring elements of answers to those three questions. I provide additional empirical evidence on the effects of cycles on productivity growth; this tends to answer question 2. I investigate whether the effects are related to whether fluctuations are more or less *transitory*. This tends to answer question 3. And I test whether Research and Development may be included in those activities which are governed by intertemporal substitution effects. This tends to answer question 1. The answers, however, hardly settle the debate.

3. Methodology

The OC theory emphasizes the role of intertemporal substitution between direct production activities and productivity-improving activities along the business cycle. Its main implications may therefore be spelled out as follows:

(i) The immediate effects of a recession is to raise the level of productivity-improving activities and accordingly total factor productivity growth.
(ii) This effect is stronger when the recession is more transitory; a permanent recession should have no effect on productivity growth.
(iii) The long-run effect of a recession on the level of productivity is positive. Whether it is higher or lower when the recession is more transitory is a priori unclear: on the one hand, reorganization proceeds at a faster speed; on the other hand it occurs for a shorter time.
(iv) The amplitude and frequency of macroeconomic fluctuations typically

have an effect on long-run growth, although the direction of these effects is ambiguous.

In order to clarify the analysis, it is useful to keep in mind a simple 'benchmark' model which will serve as a basis for my empirical strategy. I am not going to directly test this model because it is too restrictive. It will be useful, however, to introduce the transitory and permanent component indices I am going to use.

The model is a linear version of an equation derived in Aghion and Saint-Paul (1991, p. 39). It consists of the following condition:

$$v_t = \alpha\left(n_t - (1-\lambda) \sum_{i=0}^{+\infty} \lambda^i \mathrm{E}_t n_{t+i} \right), \tag{1}$$

where v_t is the 'speed of reorganization', i.e. that part of total factor productivity growth which is due to 'productivity improving activities', n_t is an indicator of the business cycle (which will also be labeled 'aggregate demand'), and $\lambda \in [0,1]$ is an appropriate 'discount factor'. The term $(1-\lambda)\sum_{i=0}^{+\infty} \lambda^i \mathrm{E}_t n_{t+i}$ represents the 'permanent component' of n_t. If n_t follows a white noise, it is equal to $(1-\lambda)n_t$; if n_t follows a random walk, it is equal to n_t. Eq. (1) then states that v_t is proportional to the transitory component of the business cycle (the difference between n_t and its permanent component). The OC approach postulates that $\alpha < 0$. If one denotes by $\hat{n}_{t+i}$ (resp. $\hat{v}_{t+i}$) the impulse response of n_t (resp. v_t) to an innovation in demand, then eq. (1) may be rewritten

$$\hat{v}_t = \alpha\left(\hat{n}_t - (1-\lambda) \sum_{i=0}^{+\infty} \lambda^i \hat{n}_{t+i} \right) \tag{2}$$

or

$$SR = \hat{v}_t / \hat{n}_t = \alpha . T(\lambda), \tag{3}$$

where $T(\lambda) = 1-(1-\lambda)\sum_{i=0}^{+\infty} \lambda^i \hat{n}_{t+i}/\hat{n}_t$ is an index of the transitory nature of demand fluctuations,[4] and SR stands for the short-run effect of a demand shock on productivity growth. Eq. (3) states that the immediate effect of a unit demand shock on v_t[5] is proportional to $T(\lambda)$ with a negative coefficient. (ii) holds, and (i) does provided $T(\lambda)$ is positive, which, as suggested by the results below, is typically the case.

[4]This index may be criticized on the grounds that the weights are decreasing in i, by contrast with the 'mean lag' indicator. It is, however, the appropriate index, since it is implied by eq. (1), and is equal to one minus 'permanent income'. If n follows an AR1 with autoregressive coefficient ρ, then $T(\lambda) = \lambda(1-\rho)/(1-\lambda\rho)$, implying $\partial T/\partial \rho < 0$.

[5]Where 'unit' is defined as an innovation which has an immediate effect on n_t equal to 1.

This linear model also puts more structure on the long-run effects than (iii) suggests, while it has nothing to say about (iv). Concerning the long-run effect of a unit demand innovation, it is equal to

$$LR=\left(\sum_{i=0}^{+\infty} \hat{v}_{t+i}\right)\Big/\hat{n}_t=\left(\sum_{i=0}^{+\infty} \alpha\left(\hat{n}_{t+i}-(1-\lambda)\sum_{j=0}^{+\infty} \lambda^j \hat{n}_{t+j+i}\right)\right)\Big/\hat{n}_t$$

$$=\alpha\sum_{i=0}^{+\infty}\left(1-(1-\lambda)\sum_{k=0}^{i}\lambda^k\right)\hat{n}_{t+i}/\hat{n}_t=\alpha\sum\lambda^{i+1}\hat{n}_{t+i}/\hat{n}_t=\alpha\lambda P(\lambda), \tag{4}$$

where $P(\lambda)=(1-T(\lambda))/(1-\lambda)$ is an index of persistence of aggregate demand fluctuations. The model therefore predicts that as long as $T(\lambda)<1$, which is also typically true, then a negative demand innovation has a positive long-run impact on the level of productivity: it also implies that this impact is larger when fluctuations are more persistent, thus getting rid of the ambiguity stated in (iii).

Concerning (iv), the model predicts that the characteristics of the stochastic process $\hat{n}_t$ have no effect on the average level of productivity growth: this is due to the linear structure of the model and the fact that innovations are on average equal to zero. In reality, however, linearity does not hold. Aghion and Saint-Paul (1991) study a model where the level of demand is constrained to be nonnegative. The model predicts that the variance of demand innovations is likely to have a *positive* effect on long-rung growth, while the frequency of fluctuations has an ambiguous, but non-zero, effect on long-run growth.

The above discussion suggests the following methodology:

1. Identify and characterize the business cycle.
2. Identify the impulse response of productivity growth to a business cycle innovation [(i) and (iii)].
3. Compute the correlation between the short-run response of productivity growth to a demand innovation and a transitory component index $T(\lambda)$ as computed from step 1 [(ii)]. Also, compute the correlation between the long-run effect of a demand innovation on productivity growth and the persistence index $P(\lambda)$.
4. Compute the correlation between the average growth rate and the amplitude and frequency of fluctuations as computed from step 1 [(iv)].

Decades of theoretical and empirical research have been devoted to the first step. Traditionally, a deterministic trend was subtracted from output and the residual was considered as the 'business cycle'. More recently this approach, as well as the distinction between 'trend' and 'cycle' itself, has come under attack. If the trend is defined as the level of productive capacity and the cycle as the degree of tension in the use of this capacity, then there is no reason to believe that the trend is deterministic and the cycle stochastic. Real business cycle theorists have forcefully pointed out [e.g., Kydland and Prescott (1982)] that 100% of fluctuations may be due to the 'trend'. Once it is recognized the trend is stochastic, there is an infinite number of ways to decompose a series between a stochastic trend – with permanent shocks – and a stochastic cycle – with transitory shocks [see Beveridge and Nelson (1981), Nelson and Plosser (1982), Campbell and Mankiw (1987)].

More recently, Blanchard and Quah (1989) have popularized a bivariate, semi-structural approach to this decomposition. They estimate a VAR in the change in output and the unemployment rate and show that under the assumption that 'demand disturbances' – or the 'cycle' – have no long-run effect on output, it is possible to recover the trend and cycle innovations from the VAR innovations in a unique way.[6]

The attractive feature of the Blanchard–Quah approach is that the long-run restriction is well grounded in traditional macro theory, which emphasizes the fact that business cycles come from nominal rigidities which should not prevent output from returning to the natural rate as nominal price adjustments proceed.

The problem is that macro theory has also changed; hysteresis models have appeared, which stress the idea that the 'natural rate' is itself determined by past demand shocks. Endogenous growth models, and in particular the OC models in which I am interested, come into this category. In this respect, the Blanchard–Quah identifying assumption is no longer a feature of the theory, and is incompatible with the very problem in which we are interested: the effects of business cycles on long-run growth.

One solution may be to go back to structural econometrics and use demand instruments to identify the cycle. This is implicitly what is done in Bean (1990). Instead of doing so, I follow a variant of the Blanchard–Quah method proposed by Gali and Hammour (1991). Since I extend this method below to take account of Research and Development, let us describe it briefly.

Gali and Hammour estimate a VAR in the Solow residual and the employment rate, which may then be inverted to get the VMA representation

[6]Bayoumi and Eichengreen (1992) have applied this approach to a bivariate system with the change in prices instead of unemployment, which is closer to the traditional AS/AD approach. In reality, this is not the only identifying assumption one has to make: it is also necessary to assume that demand and supply disturbances are uncorrelated; and more subtly, that they are fundamental with respect to the VAR innovations: i.e., that the Wold decomposition of these innovations contains all the information needed to recover the structural innovations. See Lippi and Reichlin (1991).

$$X_t = C(L)w_t = A(L)\varepsilon_t, \tag{5}$$

where $X_t = (\Delta s_t, n_t)'$, with Δs_t the Solow residual and n_t the employment rate, $w_t = (w_s, w_n)'$ the vector of VAR residuals, $C(L)$ the VMA obtained by inverting the VAR ($C(0) = I$), $\varepsilon_t = (\varepsilon_s, \varepsilon_d)'$ the vector of 'true' supply and demand disturbances, and $A(L)$ the maxtrix-lag polynomial describing the dynamic effects on X of these 'true' disturbances. Let $A(L) = \sum A_i L^i$ and $C(L) = \sum C_i L^i$; then one must have $w_t = A_0 \varepsilon_t$ and $A_i = C_i A_0$ for all i.[7] Therefore, as long as A_0 is identified, $A(L)$ is and one just recovers the εs from the ws through $\varepsilon = A_0^{-1} w$. Assume ε_s and ε_d are uncorrelated. Then, one must have $A_0 A_0' = \Omega$ is the variance–covariance matrix of the VAR residuals. Since this gives us three restrictions on the coefficients of A_0, an additional restriction is needed to identify A_0. Blanchard and Quah assume $[C(1)A_0]_{12} = 0$; i.e., no long-run effect of demand shocks on productivity. By contrast, Gali and Hammour go back to more traditional identifying assumptions and assume $[A_0]_{12} = 0$; i.e., no *immediate* effect of demand shocks on productivity.

This assumption sounds reasonable, because productivity-improving activities like reorganization, R & D, training within the firm, experimentation, implementation of new devices, etc., all work their way slowly and show up only progressively as an increase in the level of productivity. With the yearly data I use, it is implicitly assumed that a delay of one year is needed before these activities have real effects on productivity. This seems plausible, although the identifying assumption would be even better with quarterly data. For the purpose of cross-country comparisons, however, I need comparable data on business cycles for several countries, which are typically yearly.

This assumption amounts to saying that a demand shock is equivalent to an innovation in employment, so that A_0 does not have to be computed in order to get the effects of a demand shock on employment and productivity; these are just the impulse responses obtained from the VAR estimates. Of course, A_0 has to be computed in order to get the effects of supply shocks.

Note that this technique solves steps 1 and 2 simultaneously, since it is precisely an assumption about step 2 which allows us to solve step 1.

Although the identifying assumption seems to make sense from a theoretical point of view, how can we reconcile it with the observed procyclicality of the Solow residual? First, one may assume that this procyclicality is just due to the response of employment to productivity shocks. It is then in some sense a misuse of the word. In this case, the identifying assumption remains perfectly valid, and it indeed implies that all the contemporaneous correlation between productivity and employment inno-

[7]This comes from the fundamentaliness assumption about the ε's and the uniqueness of X's Wold representation; see the previous footnote.

vations is due to supply shocks. However, this procyclicality may also be due to mismeasurement of the residual:

- mismeasurement of the labor input (labor hoarding; procyclical effort), or:
- mismeasurement of the labor contribution (increasing returns or imperfect competition).

The second case has been emphasized by Hall (1990). In both cases, the Solow residual will be correlated with demand instruments, so that even if a demand shock has no immediate effect on true productivity growth, it will have an effect on measured productivity growth. In this case the identifying assumption is no longer valid, unless one corrects for mismeasurement. To take this phenomenon into account, I also compute the results using the following correction:[8] I first estimate a forecasting equation for three 'demand' variables: money growth, exchange rate depreciation, and the change in government consumption. For each variable z the forecasting equation is

$$z_t = m(L)z_{t-1} + n(L)y_{t-1} + \tilde{z}_t,$$

where y is the change in output. I then regress the Solow residual Δs_t on the change in the rate of capacity utilization $\Delta\rho_t$ (or alternatively average weekly hours in manufacturing) using the forecast errors $\tilde{z}_t$ as instruments. Let $\hat{\beta}$ be the regression coefficient. The corrected Solow residual is then defined as

$$\Delta\tilde{s}_t = \Delta s_t - \hat{\beta}\Delta\rho_t.$$

The use of forecast errors $\tilde{z}_t$ as instruments instead of the demand variables themselves serves two purposes. First, if aggregate demand policy reacts to past supply shocks, and if these are autocorrelated, then z_t will be correlated with the error term in the regression, but not $\tilde{z}_t$. Second, since the model predicts that past demand shocks have an effect of productivity growth, this is also likely to generate a correlation between z_t and the error term. This is why only 'unanticipated' demand variables should be used as instruments.[9]

An alternative approach to identification relies on the use of the model described in eqs. (1) to (4). For any given impulse response of employment to a demand shock, the model implies a relationship between the immediate effect of a demand shock on v_t, SR_t, and its long-run effect on total factor productivity LR_t. This restriction may be used as a means to identify A_0. In

[8]Gali and Hammour correct for labor hoarding using a cyclical elasticity of effort obtained from microeconomic survey data [in particular Fay and Medoff (1985)]. Since steps 3 and 4 of my work are based on cross-country exercises, this is inappropriate here: first, such studies are not available for all countries; second, one cannot extrapolate the coefficient for the U.S. to other countries: the coefficient probably varies across countries, due to differences in labor market regulation like firing costs, etc.

[9]As long as the instruments are valid, this correction eliminates all the sources of mismeasurement mentioned above: labor hoarding, imperfect competition and increasing returns.

fact, it is possible to show that it is possible to eliminate A_0's coefficients from the restriction and to get a second degree equation in α (see appendix). An alternative means to test the OC approach is then to compute these roots and look at their sign. This is what I do below as a robustness check on the sign of SR. However, this identifying assumption is more fragile than the Gali–Hammour one, because it relies on the geometric structure of the lead-polynomial in (1).

Once the VAR technique I just described is applied, it is possible to compute the impulse response of productivity growth to an innovation in demand, as well as the transitory and persistence indices $T(\lambda)$ and $P(\lambda)$ and the variance of demand and supply innovations.

The next step is obtained by repeating this for several countries, and look at the cross-country correlation of $T(\lambda)$ and the short-run effect of demand shocks on productivity growth. Given the identifying assumption that the immediate effect is zero, the short-run effect is defined as the one-year-ahead effect; in terms of the above model, it is assumed that $\Delta s_t = v_{t-1}$. I also look at the cross-country correlation of the long-run effect LR and the persistence index $P(\lambda)$.

I then run step 4 by regressing the average growth rate on the variance of demand innovations, σ_α, and indicators of the frequency of fluctuations like $T(\lambda)$ and the spectral density of demand innovations decomposed into a small number of frequency bands. This allows to test whether wider fluctuations are good or bad for growth, and whether fluctuations of a higher or lower frequency are good or bad for growth. I include traditional control variables like secondary enrollment and the initial output levels.

The implications of the OC approach are then, by order of importance:

- $SR < 0$,
- $LR < 0$,
- SR is negatively correlated with $T(\lambda)$ in a cross-section,
- LR is negatively correlated with $P(\lambda)$ in a cross-section,
- σ_d and the frequency indicators have significant effects on long-run growth in a cross-section.

The clearest rejection of the OC approach would occur if it was found that SR is significantly positive. In this case, the immediate effect of a recession on productivity growth would be in the opposite direction from what the OC theory predicts. The next strong implication of the theory is the second one, namely that $LR < 0$. It is less strong than $SR < 0$ because a negative innovation in demand may well lead to a systematic expansion in the future, which will then be associated with a negative effect on productivity growth.[10] Also, the short-run effects may be due to changes in the *timing* of

[10]Under the geometric lead structure of eq. (1), LR should still be negative as long as $P(\lambda) > 0$. However, this no longer holds for non-geometric lead structures.

activities, not to their amount. In this case one would have short-run effects but no long-run effects.

The predictions concerning the cross-section correlations of short- and long-run effects with $T(\lambda)$ and $P(\lambda)$ are somewhat more fragile, because they rely on the assumptions that (i) $T(\lambda)$ and $P(\lambda)$ are the appropriate indicators for the transitory and permanent component; (ii) the coefficient α is the same across countries, or at least it does not vary in a way which is systematically correlated with $T(\lambda)$ and $P(\lambda)$; and (iii) the estimated shocks are comparable across countries. Of the two predictions, the negative correlation between LR and $P(\lambda)$ is the least robust, because it heavily relies on the fact that the weights in (1) are geometric. If the relevant persistence index is a more complex lead-polynomial than $P(\lambda)$, then no clear prediction emerges. The prediction that SR is negatively correlated with $T(\lambda)$ is more robust because any good index of the transitory component is likely to be positively correlated with $T(\lambda)$. In other words, the third of the five implications listed above is a general test of the OC model, whereas the fourth one is a test of eq. (1).

Finally, the fifth prediction does not imply any direct test of the OC theory: for example, if one assumes that recessions are shorter than expansions, an increase in the frequency of recessions would raise the high-frequency components of the spectrum relative to the low frequencies. In the Aghion–Saint-Paul model, this would be associated with more growth if recessions are too infrequent to start with, but less growth if they are too frequent. Therefore there is no clear prediction. This, however, suggests that looking at the correlation between long-run growth and the stochastic properties of the business cycle is relevant.

4. Results

The vector autoregression was run using data for 22 OECD countries, 1950–1988, from the *OECD Economic Outloook* data set. In order to increase the stationarity of the series, a time trend was included in the employment equation and a post-1974 dummy in the productivity growth equation. The VAR was run both for corrected and non-corrected series. Two lags were used.

Table 1 shows the implied long-run and short-run responses of productivity to demand shocks for the non-corrected series. The result, in line with the previous evidence in Bean and Gali–Hammour, supports the OC approach. The short-run effect is negative in 15 countries and significantly so in 7 of these 15 countries; by contrast, it is never significantly positive. The long-run effect is significantly negative in 5 countries and never significantly positive. One has to deplore the very imprecise estimate of the long-run effect in the U.S., which is at variance with the Gali–Hammour results. The orders

Table 1

Effect on productivity of a demand innovation.[a]

Country	Short-run (SR)	Std. error	Long-run (LR)	Std. error
Australia	−0.70	0.26	−0.82	0.47
Austria	−0.23	0.19	−1.37	0.77
Belgium	−0.24	0.24	0.10	0.53
Canada	−0.32	0.21	−0.23	0.27
Denmark	−0.48	0.21	−0.76	0.34
Finland	−0.39	0.25	−1.22	0.88
France	−0.32	0.15	0.14	0.37
Germany	−0.47	0.17	−0.64	0.37
Iceland	0.20	0.58	−0.83	1.01
Ireland	0.20	0.28	−0.02	0.40
Italy	−0.01	0.23	0.16	0.30
Japan	0.06	0.46	0.23	1.42
Luxembourg	0.00	0.51	−0.09	1.47
Netherlands	−0.30	0.20	−0.31	0.46
Norway	−0.23	0.28	−1.44	1.19
New Zealand	−0.24	0.24	0.59	0.60
Portugal	0.70	0.18	3.38	7.27
Spain	−0.02	0.17	−0.33	0.49
Sweden	0.02	0.32	0.41	0.45
Switzerland	0.03	0.16	−0.25	0.14
U.K.	−0.52	0.18	−0.81	0.17
U.S.	−0.92	0.23	−5.58	16.3

[a]Non-corrected data, 50–88; effect of a demand innovation implying a unit contemporaneous increase in employment. Data from *OECD Economic Outlook*.

of magnitude are plausible, except perhaps for the (very imprecise) U.S. long-run effect.

Unfortunately, these results disappear when the corrected data are used: no significant effects of demand shocks on productivity growth emerge, whether in the short or in the long run. This is both disappointing and puzzling, since one would have tended to believe that the correction should increase any counter-cyclicality of the Solow residual (because it takes out a component positively correlated with demand shocks out of it). This turns out not to be the case, and may be due to the poor quality of instruments: in fact the $\hat{\beta}$ coefficient in the correction regression is often imprecisely estimated and/or negative.[11]

I therefore proceed with the results for non-corrected data only. First, I use the alternative identifying restriction which links the long-run and

[11]The correction coefficient $\hat{\beta}$ is significant for six countries: Australia, Finland, France, Spain, U.K. and U.S. For Spain it is implausibly large. For the other five it is between 1.2 and 1.3. For other countries it is often too large and/or negative. The corrected data imply positive effects of demand on growth for the U.S. (instead of negative), and no significant effects for the rest except for a significant negative short-run effect for Australia.

Table 2

Estimates of α using eqs. (1) and (4).[a]

Country	α_1	α_2
Australia	−0.416	−1.49
Austria	−0.31	20.22
Belgium	0.03	−3.34
Canada	−0.12	−8.47
Denmark	−0.46	−5.65
Finland	−0.35	−37.5
France	0.04	486.6
Germany	−0.59	−0.71
Iceland	−0.55	−5.69
Ireland	−0.006	−3.02
Italy	0.05	18.7
Japan	0.055	−444.6
Luxembourg	−0.02	−5.1
Netherlands	−0.17	−8.25
Norway	−0.18	−4.52
New Zealand	0.74	−3.36
Portugal	1.00	−3.75
Spain	−0.04	−7.51
Sweden	0.17	58.77
Switzerland	−0.31	−1.81
U.K.	−0.32	−2.59
U.S.	*	*

[a]α_1 and α_2: roots of eq. (A.6); α_1 = most plausible root (for Germany, both roots are equally plausible); for the U.S., negative determinant.

short-run effects of demand on productivity through eqs. (1) and (4). The two estimates of α which it gives are reported in table 2. In most cases, only one estimate is plausible in magnitude so that the restriction in fact allows to fully identify the model; furthermore, α is quite often negative which, like the results of table 1, supports the OC approach. Of course, in this case, the effects on *LR* are, by assumption, exactly in the same direction as those on *SR*.

I now proceed with the cross-country analysis, starting with the correlation of $T(\lambda)$ and SR for non-corrected data. I report the results for $\lambda = 0.9$, but they are reasonably robust to different values of λ between 0.8 and 1.

As illustrated in fig. 1, there is a negative cross-section correlation between $T(0.9)$ and SR. This is in line with the prediction of the OC approach. This correlation is, however, rather weak, as suggested by the following regression (t-statistics in parentheses):

$$\underset{(0.15)}{SR = 0.0329} - \underset{(-1.2)}{0.41} \cdot T(0.9), \quad R^2 = 0.07.$$

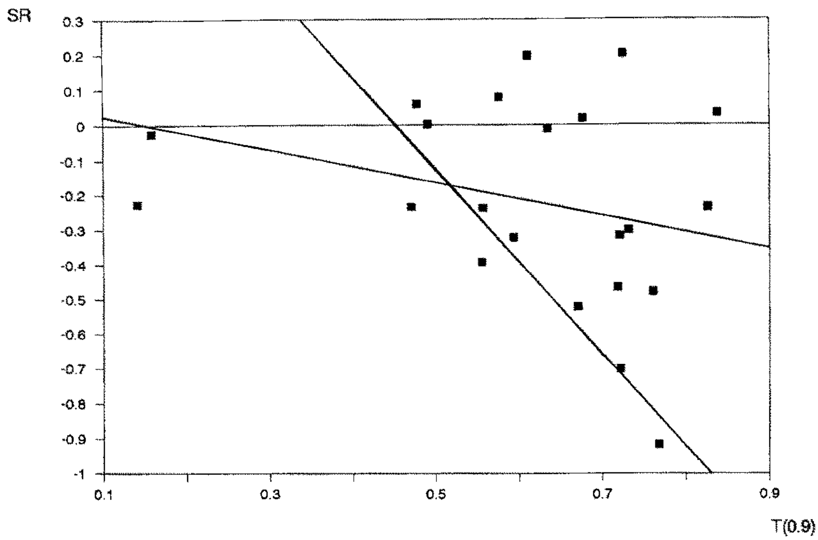

Fig. 1

When observations are weighted according to population, the regression is significantly more supportive of the OC approach:

$$SR = 1.27 - 2.77 \cdot T(0.9), \quad R^2 = 0.94.$$
$$(5.62)(-8.61)$$

The two regression lines are represented in fig. 1.[12]

By contrast with the moderate success in testing the negative correlation between SR and $T(0.9)$, all attempts to get a significant correlation between LR and $P(0.9)$ failed. This suggests, as discussed above, that the geometric formulation is too restrictive.

I now turn to the growth regressions. The results are as follows: first, the effect of the variance of demand innovation on long-run growth tends to be negative, which tends to contradict the Aghion–Saint-Paul model. The coefficient is, however, quite insignificant, so that whether there is any effect of volatility on long-run growth remains an open question.

Table 3 presents regressions of long-run growth on frequency indicators. The frequency indicators that were used were:

– $T(0.9)$: the transitory index;

[12]It may be argued that the weighted regression gives too much weight to the U.S.: this explained the strong negative correlation. The truth probably lies between the results of the weighted and unweighted regressions.

Table 3

Growth regressions.[a]

Independent variables	Dependent variable: *GYD* (1)	(2)	(3)	(4)	(5)	(6)
C	3.69	2.44	3.19	2.48	2.47	2.91
	(5.06)	(3.56)	(5.00)	(3.35)	(5.07)	(4.87)
YD65	−0.05	−0.086	−0.05	−0.40	−0.49	−0.43
	(−0.95)	(−1.71)	(−1.01)	(−6.32)	(−10.5)	(−13.8)
SE	0.01	−0.00	0.00	0.05	0.05	0.05
	(0.88)	(−0.27)	(0.27)	(5.83)	(4.78)	(6.03)
T(0.9)	−2.42	–	–	−0.65	–	–
	(−2.65)			(−0.52)		
SP1	–	1.95	–	–	1.59	–
		(2.68)			(1.71)	
SP4	–	–	−4.17	–	–	−3.94
			(−3.09)			(−1.85)
R^2	0.43	0.44	0.49	0.98	0.98	0.98
Weight	None	None	None	*POP*	*POP*	*POP*

[a]20 observations (Iceland and Luxembourg excluded); *SE* = secondary enrollment; *YD65* = Real 1965 income. 1987 dollars; *C* = constant: *GYD* = real GDP per capita growth, annual, 1965–1987; *POP*: population, 1987; *T*(0.9), *SP4*, *SP1*, defined in text. *t*-statistics in parentheses. Data for *SE*, *POP*, *GYD* and *YD65* from *World Development Report*, 1989.

- *SP4*: the proportion of the variance of demand shocks explained by those frequency components whose period is between 2 years and 4 years.
- *SP1*: the proportion of the variance of demand shocks explained by those frequency components whose period is more than 16 years.

The reason why *SP1* and *SP4* come into the regression rather than other components of the spectrum is simply that those are the most significant.

A stylized fact seems to emerge from table 3, namely that countries where fluctuations are more transitory have lower rates of growth: both *T*(0.9) and *SP4* have negative impact on growth, while *SP1* has a positive impact on growth. This is somehow weakened when observations are population-weighted. Figs. 2, 3, and 4 show that even when conditioning variables are not included, this regularity is quite strong.

5. Implications for the trend vs. cycle decomposition

One interesting aspect of the above approach is that it sheds a new light on the trend vs. cycle decomposition of output. As I pointed out above, an important debate in the eighties was whether part of what was traditionally considered as the 'cycle' was in fact the response to productivity shocks, i.e. innovations in the trend. The endogenous growth models I discussed above emphasize the fact that part of 'trend' – i.e. the Solow residual – is in fact

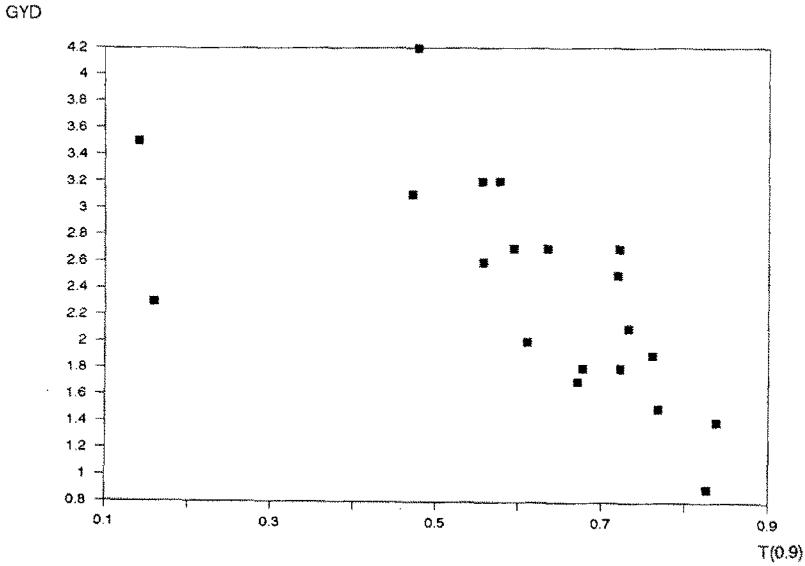
GYD
4.2
4
3.8
3.6
3.4
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.1
0.3
0.5
0.7
0.9
T(0.9)

Fig. 2

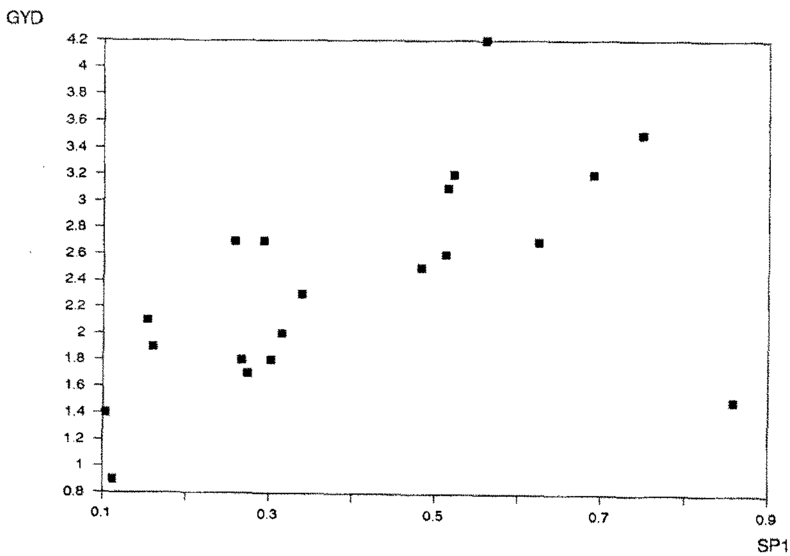
GYD
4.2
4
3.8
3.6
3.4
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.1
0.3
0.5
0.7
0.9
SP1

Fig. 3

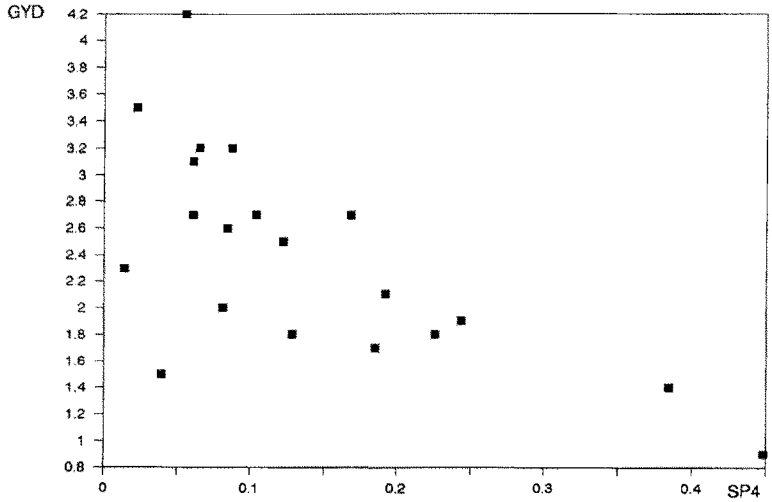

Fig. 4

due to the response of productivity growth to aggregate demand shocks. It is therefore useful to compute which part of the variance of the Solow residual is actually due to demand shocks. Another interesting question is whether the above VAR approach gives more weight to demand shocks than the Blanchard–Quah identifying assumption.

Table 4 shows the variance decomposition of the Solow residual and the employment rate for the major OECD countries.

As table 4 shows, most of the variance of the Solow residual is due to supply shocks, while most of the variance of employment is due to demand shocks. One exception is the U.S. in the long run, but this should not be taken too seriously in light of the large standard errors attached to the long-run effects in this case.

The proportion of the variance of the Solow residual explained by demand shocks is not always negligible, however, since it is around 10% for France and the U.K. and 20% for Germany and the U.S. The very low share of (short-run) employment fluctuations explained by supply shocks is somewhat puzzling, as is the fact that this share is higher in America than in Europe (one might have expected the contrary from the fact that European unemployment is more persistent).

6. Evidence from R & D data

One important question is whether research and development may be included in the list of activities which go up in recessions. R & D expenditures are traditionally believed to be 'pro-cyclical' [see for example Hall and

Table 4

Variance decomposition of productivity and employment.[a]

	Horizon:			
	0 years	1 year	5 years	$+\infty$
1. Productivity:				
Canada	0.0	3.8	4.0	4.0
France	0.0	10.0	11.0	11.5
Germany	0.0	19.8	19.6	19.7
Italy	0.0	0.0	0.3	0.3
Japan	0.0	0.0	0.1	0.1
U.K.	0.0	11.3	11.6	11.6
U.S.	0.0	27.3	26.2	16.8
2. Employment:				
Canada	71.4	63.6	62.0	62.0
France	89.8	85.5	88.4	89.1
Germany	99.4	93.4	80.9	79.5
Italy	99.9	98.5	95.8	95.7
Japan	89.3	89.0	88.9	88.9
U.K.	86.2	67.2	55.2	55.1
U.S.	60.9	46.3	29.2	16.8

[a]Proportion of the variance of employment and the Solow residual at $t+i$ explained by demand innovations, in percent, conditional on the information available at $t-1$, for $i=0, 1, 5, +\infty$.

Mairesse (1991)], but this may well be due to the response of R & D expenditures to supply shocks rather than demand shocks. It is therefore interesting to isolate these two effects.

To do so, I extend the above VAR methodology in the following way: let now $X_t=(\Delta s_t, n_t, \Delta R_t)'$, where, as above, Δs stands for the Solow residual, n for the employment rate and ΔR for an indicator of R & D activity. Let $X_t=B(L)w_t$ be X's VMA representation, and $X_t=A(L)\varepsilon_t$ be its representation in terms of the true disturbances, where $\varepsilon=(\varepsilon_s, \varepsilon_d, \varepsilon_r)'$ is a vector of supply, demand, and R & D disturbances. An R & D disturbance may be thought of, for example, as a change in the tax treatment of R & D expenditures. Then one must again have $A(L)=B(L)A_0$. Given that A_0 is a 3×3 matrix with $A_0A_0'=\Omega$, one needs three restrictions to identify the true disturbances. The three identifying assumptions I use are the following:

1. A demand shock has no immediate effect on the Solow residual: $A_{12}=0$.
2. An R & D shock has no immediate effect on the Solow residual: $A_{13}=0$.
3. The long-run effect of a demand shock on total factor productivity is the same as its long-run effect on R & D: $[B(1)A]_{12}=[B(1)A]_{32}$.

The first two assumptions rest on the above discussed idea that productivity-improving activities show up only gradually in the level of total factor productivity. Assumption 3 amounts to assuming that a demand shock has no long-run effect on the R & D/output ratio. In other words, if only demand shocks existed, there would be a cointegrating relation between R & D and output. This is not exempt from criticism, but is reasonable: the R & D/output ratio is a proxy for the *return* to R & D activities. One does not see why demand shocks which eventually vanish should affect this return in the long run. By contrast, R & D shocks like changes in the tax treatment of R & D will affect this return, as well as technology shocks if they have a differential impact on productive technology and R & D technology, or if they radically change the prospects for innovation.

As a check to robustness, I also try another set of identifying restrictions, namely replacing (3) by the assumption that the immediate effect of a demand shock on R & D is zero. This may be justifiable if there is some slugglishness associated with R & D activities.

The VAR was run for the major OECD countries with corrected data; OECD data on R & D expenditures and patents were used. Given that they start in 1970, I put only one lag in the VAR.

The results are reported in table 5. Several remarks have to be made. First, when expenditures are used as the R & D variable, there is very little evidence of any pro- or counter-cyclical behavior: the immediate impact of a demand shock is sometimes positive, sometimes negative, and always very insignificant. Second, when patents are used, the estimates are more precise, but extremely sensitive to the identifying assumption which is used. If one believes in the first identifying assumption, then the picture which emerges is that of a strong counter-cyclical behavior of patents – in line with the theory.

7. Conclusion

This paper has attempted to analyze the growth effects of business cycles. The business cycle was identified using now popular multivariate techniques. Although the results are at best mixed, several conclusions emerge:

1. There is more empirical support for the opportunity cost approach than one might have expected: the results very often imply a negative immediate effect of demand innovations on productivity growth. This is, however, less true in the long run.
2. In a cross-section of countries, the effect of demand on productivity growth is stronger when demand fluctuations are more transitory, which supports the idea that intertemporal substitution is important.
3. The response of productivity growth to demand shocks explains a small, but non-negligible share of the variance of the Solow residual: 10–20%.
4. There is no evidence of an effect of the volatility of demand shocks on long-run growth.

Table 5

Effect of demand innovations on R & D.[a]

	A			B	
	Inst.	1 year	Cumul.	1 year	Cumul.
1. Expenditures					
U.S.A.	0.53	0.22	0.2	−0.09	−1.31
	(1.07)	(0.52)	(1.1)	(0.24)	(4.3)
Germany	−0.10	−0.14	−0.16	−0.1	0.07
	(0.36)	(0.39)	(1.19)	(0.29)	(0.7)
France	0.00	−0.12	−0.13	−0.12	−0.14
	(0.26)	(0.16)	(0.07)	(0.12)	(0.22)
Japan	−1.65	0.56	−0.88	0.86	1.19
	(2.88)	(1.59)	(0.93)	(1.85)	(4.05)
Italy	1.63	−0.07	−0.79	−0.45	−2.72
	(3.07)	(0.63)	(0.63)	(0.83)	(3.37)
Canada	0.45	0.14	0.03	−0.07	−0.69
	(3.82)	(1.22)	(0.68)	(0.56)	(6.72)
2. Patents					
U.S.A.	−6.16	2.87	0.28	0.89	4.11
	(1.29)	(1.57)	(1.56)	(0.25)	(5.49)
U.K.	−2.47	−0.81	−2.66	−0.41	−0.24
	(2.19)	(0.48)	(2.46)	(0.24)	(2.12)
Germany	−1.11	−0.39	−0.36	−0.17	0.76
	(0.22)	(0.21)	(0.28)	(0.11)	(0.68)
France	2.09	−1.40	−0.02	−0.87	−1.85
	(0.92)	(0.84)	(0.17)	(0.27)	(0.86)
Japan	−3.33	1.05	−0.23	2.1	6.02
	(1.66)	(0.90)	(0.57)	(1.14)	(4.32)
Italy	1.71	0.49	−0.72	−0.77	−6.30
	(1.75)	(0.92)	(0.73)	(0.76)	(5.38)
Canada	−8.04	4.61	−1.44	1.61	4.54
	(0.76)	(2.13)	(0.64)	(0.56)	(4.04)

[a]Standard errors in parentheses. Panel A: identifying assumptions 1, 2, and 3 in the text. Panel B: 3 replaced with the assumption that demand has no contemporaneous effect on R & D. Data on expenditure not available from this source for the U.K. Data sources for R & D: *OECD Main Science and Technology Indicators* (1991). Expenditure series: BERD, p. 26, table 21–22. Patents: domestic patent applications, table 72, p. 53.

5. High frequency components of fluctuations seem to have a negative effect on long-run growth, while low frequency components tend to have a positive effect. This seems to be a rather robust stylized fact. It is therefore important for future research to document this fact more widely (how robust is it to alternative identifying assumptions?), and to provide theoretical foundations for it. Although it may be consistent with OC models like Aghion–Saint-Paul, completely different explanations may be envisaged.

6. There remains very little evidence of any pro- or counter-cyclical behavior of R & D once one tries to distinghish between demand and supply shocks (this may be just due to the poor quality of aggregate R & D data).

Appendix: Using the model to identify structural innovations

Using the notations of section 3, the immediate effect of a demand innovation on productivity growth is given by

$$\hat{v}_t = [C_0 A_0]_{12} = [A_0]_{12}; \tag{A.1}$$

the i-period ahead effect of a demand innovation on employment is given by

$$\hat{n}_{t+i} = [C_i A_0]_{22}. \tag{A.2}$$

Eq. (1) may therefore be rewritten

$$[A_0]_{12} = \alpha[(I-(1-\lambda)C(\lambda))A_0]_{22}. \tag{A.3}$$

Eq. (4) may be written as

$$[C(1)A_0]_{12} = \alpha\lambda[C(\lambda)A_0]_{22}. \tag{A.4}$$

Eq. (A.4) is equivalent to

$$[A_0]_{12} = \alpha \frac{1-(1-\lambda)[C(\lambda)]_{22}}{1+\alpha(1-\lambda)[C(\lambda)]_{21}} [A_0]_{22}. \tag{A.5}$$

Substituting (A.5) into (A.3) and multiplying by the denominator of (A.5) yields, after simplification,

$$\begin{aligned}\{\lambda[C(\lambda)]_{21}\}\alpha^2 + \{&\lambda[C(\lambda)]_{22} - (1-\lambda)[C(\lambda)]_{21}[C(1)]_{12} \\ &+ (1-\lambda)[C(\lambda)]_{22}[C(1)]_{11} - [C(1)]_{11}\}\alpha \\ &- \{[C(1)]_{12}\} = 0.\end{aligned} \tag{A.6}$$

This is a degree 2 equation which may be solved once the VAR coefficients are estimated.

References

Aghion, P. and G. Saint-Paul, 1991, On the virtues or bad times, CEPR Working paper no. 578 (CEPR, London).

Amber, S., E. Cardia and L. Phaneuf, 1991, Contrats de salaire, croissance endogène et fluctuations, Working paper (Université du Québec à Montréal, Montreal).

Bayoumi, T. and B. Eichengreen, 1992, Shocking aspects of European monetary unification, CEPR Working paper no. 643 (CEPR, London).

Bean, C., 1990, Endogenous growth and the pro-cyclical behaviour of productivity, European Economic Review 34, 355–363.

Bean, C. and C. Pissarides, 1993, Unemployment, consumption and growth, European Economic Review 37, this issue.
Beveridge, S. and C.R. Nelson, 1981, A new approach to decomposition of economic time series into permanent and transitory components with particular attention to measurement of the 'business cycle', Journal of Monetary Economics 7, no. 2, 151–174.
Blanchard, O.J. and D. Quah, 1989, The relative importance of demand and supply disturbances, American Economic Review, Sept.
Caballero, R., 1993, Demand, unemployment, and growth: Comment, European Economic Review 37, this issue.
Caballero, R. and M. Hammour, 1991, The cleansing effect of recessions, NBER Working paper no. 3922, Dec. (NBER, Cambridge, MA).
Campbell, J. and N.G. Mankiw, 1987, Are output fluctuations transitory?, Quarterly Journal of Economics, Nov., 857–880.
Davis, S. and J. Haltiwanger, 1989, Job creation, job destruction, and job reallocation over the cycle, NBER Macroeconomics Annual (NBER, Cambridge, MA).
Dellas, H., 1991, Stabilization policy and long-run growth: Are they related?, Mimeo. (University of Maryland, College Park, MD).
Fay, J. and J.L. Medoff, 1985, Labor and output over the business cycle: Some direct evidence, American Economic Review.
Gali, J. and M. Hammour, 1991, Long-run effects of business cycles, Mimeo. (Columbia University, New York).
Hall, R.E., 1990, Invariance properties of the Solow residual, in: P.A. Diamond, ed., Growth, productivity, unemployment (MIT Press, Cambirdge, MA).
Hall, R.E., 1991, Recessions as reorganizations, NBER Macroeconomics Annual (NBER, Cambridge, MA).
Hall, B. and J. Mairesse, 1991, Exploring the relationship between R and D and productivity at the firm level in French manufacturing, Mimeo. Dec.
Lippi, M. and L. Reichlin, 1991, VAR analysis, non-fundamental representation, Blaschke matrices, Mimeo. (OFCE, Paris) Oct.
King, M. and M. Robson, 1990, Endogenous growth and the role of history, Mimeo. (LSE, London).
Kydland, F. and E. Prescott, 1982, Time to build and aggregate fluctuations, Econometrica.
Nelson, C. and C. Plosser, 1982, Trends and random walks in macroeconomic time series, Journal of Monetary Economics.
Nickell, S., S. Wadhwani and M. Wall, 1992, Productivity growth in U.K. companies, 1975–1986, European Economic Review 36, no. 5, June, 1055–1085.
OECD, 1991, Main science and technology indicators (OECD, Paris).
OECD, 1992, Economic outlook (OECD, Paris).
Saint-Paul, G., 1991, Productivity growth and unemployment in OECD countries, Working paper no. 91-09 (DELTA, Paris).
Schumpeter, J.A., 1939, Business cycles (McGraw-Hill, New York).
Schumpeter, J.A., 1950, Capitalism, socialism and democracy (Harper and Row, New York).
Shleifer, A., 1986, Implementation cycles, Journal of Political Economy.
Stadler, G.W., 1990, Business cycle models with endogenous technology, American Economic Review 80, 763–778.
World Bank, 1989, World development report (World Bank, Washington, DC).